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Project Title:	Human Factors Behavioral Performance Risk Characterization Research for Artemis II		
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Comments:			
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Flight Program:			
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Key Personnel Changes/Previous PI:	Added: Sydney Begerowski, PhD, KBR (co-I)		
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Task Description:	Artemis II will mark the first time NASA astronauts go beyond low-Earth orbit (LEO) since the Apollo era, and the first astronauts heading into space in the Orion vehicle. As such, it provides a critical opportunity to refine our understanding of the likelihood and consequences associated with the Behavioral Medicine (BMed), Team, Human System Integration Architecture (HSIA), and Sleep Risks, and prepare for future Moon and Mars missions. The overarching goal of our research is to utilize Artemis II data to further define the likelihood and consequences of these risks, and to create a research infrastructure that can be expanded to include future Artemis missions. We propose a 3-phase research effort. In Phase I, we will use summaries of existing research (e.g., evidence books and Directed Acyclic Graphs/DAGs), literature reviews, and subject matter expert (SME) input to identify and conceptualize key performance metrics, contributing factors, and BMed, Team, HSIA, and Sleep risk constructs related to performance decrements. We will describe how the constructs are expected to arise in spaceflight, and conduct a review that summarizes how metrics (e.g., behaviors and trace data, words and linguistics, and physiological data) derived from data streams available in Artemis II (e.g., audiovisual data) can serve as indicators of these constructs for in-mission measurement during Artemis. The Phase I effort will result in a finalized pre- and post-mission protocol for Artemis II, along with a measurement and coding scheme for in-mission Artemis II data. Phase II includes data collection from the Artemis II mission. Phase III will include data processing, data analysis under a Bayesian framework, coding, depiction, analysis, and report writing.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Artemis II provides a critical opportunity to refine our understanding of the likelihood and consequences associated with BMed, Team, HSIA, and Sleep Risks, and prepare for future Moon and Mars missions. Two significant constraints shape the research methodology. First, there is currently no in-mission crew time available to complete measures. In-mission data will need to be collected unobtrusively from available data streams (e.g., audiovisual, existing records such as schedules, and actigraphy). Second, Artemis II is anticipated to be a crew of 4 astronauts in flight for 10 days followed by additional crewed Artemis missions. This creates a scenario where there is important data for understanding risk characterization; however, there will be a small sample size from Artemis II, and risk characterization will need to be updated over time as additional crews fly in later Artemis missions.
Research Impact/Earth Benefits:	Our research program and associated deliverables will substantially reduce NASA's Human Research Program Behavioral Medicine (BMed), Team, Human Systems Integration Architecture (HSIA), and Sleep Risks and inform gaps in knowledge related to future Moon and Mars missions. Our primary contributions will be in advancing less obtrusive methodologies for monitoring and measuring risk metrics; examining potential BMed, Team, HSIA, and Sleep risks in a new vehicle, with new mission scenarios and procedures; and characterizing risks beyond LEO as multiple stressors converge. The advances in less obtrusive methodologies may have broader application to other private and public sector organizations interested in team dynamics, well-being, sleep, and human factors. Given that Artemis II will be the first human spaceflight in Orion, these data will be particularly valuable for informing modifications necessary for future Moon and Mars missions. Finally, Artemis II data will be the first systematic data collection related to these risks beyond LEO and allow for the potentially additive or synergistic effect of multiple stressors on sleep, and psychological and team health as the crew interacts with new vehicle and system interface deigns, as well as mission scenarios.
Task Progress:	Over the past year, the Behavioral Health and Performance Laboratory and collaborators from the Fatigue Laboratory and Human Systems Integration Architecture (HSIA) concluded Phase I of the project. The research team has identified key constructs pertaining to the BMed, Teams, Sleep, and HSIA risk areas as well as performance indicators likely to be most relevant to Artemis II mission. Alongside these efforts, the team consulted with relevant subject matter experts (SMEs) and conducted an interdisciplinary literature review to understand current methods and tools for unobtrusive measurement approaches. Using results from the review and the current understanding of Artemis data-collection capabilities, the research team developed a codebook for in-mission data collection that can be used to characterize the risks in an unobtrusive manner. Behavioral medicine risk Following a review of spaceflight and spaceflight analog evidence, the team identified positive and negative affect, coping, cognitive performance, and cognitive workload as the five core constructs for the behavioral medicine risk with 10 subconstructs. Pre- and post-mission data collection entails survey measurement, crew debriefing, and administration of the cognition battery. In-mission data collection heavily involves the use of natural language processing (NLP) tools such as LIWC and STRESSNet to analyze text-based files generated from audio recordings. Team risk Constructs under the Team risk were evaluated from the Team Risk deliverable presented to Human Subjects Review Board (HSRB) in 2021. Key constructs include team cognition, team functionality and performance, and interpersonal relations with 10 subconstructs for risk characterization in Artemis missions. Pre- and post-mission data collection utilizes debriefs and surveys, employing measures from the Human Factors & Behavioral Performance (HFBP)-Exploration Measures suite when appropriate. Assessment of constructs in-mission largely relies on computer-aided text analysis (CATA) approaches. Alternative methods under the team risk encompass behavioral coding methods, which require access to audio-visual data for researchers to indicate behavioral indicators of each construct. Sleep risk Drawing from foundational sleep literature and subject matter expert (SME) consultation, we identified sleep quality, fatigue, workload, and sleep inertia as key constructs for Sleep risk characterization in Artemis missions. These constructs were selected given the likelihood of being impacted by contextual features of Artemis missions (e.g., small crew sleeping quarters, unique operational tempo). In-mission measurement triangulated common methods in spaceflight (e.g., actigraphy) with other data sources (e.g., noise dosimeter) to better refine our understanding of the core constructs. HSIA risk The list of core HSIA constructs was refined to include human-systems resilience, vehicle maintainability and

	repairability, system usability, and information accessibility to better characterize the HSIA risk given constraints introduced by beyond-low Earth orbit missions such as Artemis (e.g., increased communication delays, reduced ground support). Post-mission methods include crew debriefing and surveys (e.g., NASA-Modified System Usability Scale). In-mission methods behavioral coding based on audio-visual recordings and SME evaluation using mission artifacts (e.g., mission timelines, corrective action reports, and systems data). Performance metrics To characterize performance decrements due to BMed, Team, HSIA, and Sleep Risks, we refined our measurement of performance within the Artemis mission. We intend to assess performance in terms of effectiveness (i.e., attainment of mission objectives and goals) and efficiency (i.e., work accomplished relative to the resources available). These constructs will be further refined based on SME input and mission evaluation documents. Furthermore, we will capture performance behaviors using a behaviorally anchored rating scale developed for use in extreme teams. Trained raters will indicate crew performance using this scale after observing audio-visual recordings of crew in-mission.
Bibliography Type:	Description: (Last Updated: 08/01/2025)
Conference Materials (Downloadable)	Bell ST, Begerowski SR, Anderson SA, Dev SI, Landon LB, Khader AM, Dickson P, Vera A, Flynn-Evans E, Young M, Garret J. "Human factors behavioral performance risk characterization research for Artemis II." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024 IWS 2024 Artemis II research.pdf